

Quantum Processes and Computation Mini-Project

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Question 1

The labelled ZX -rules shown below [2] for $\alpha, \beta \in [0, 2\pi)$ and $a \in \{0, 1\}$ are as taught in lectures [1], and will be used to justify the equations below.

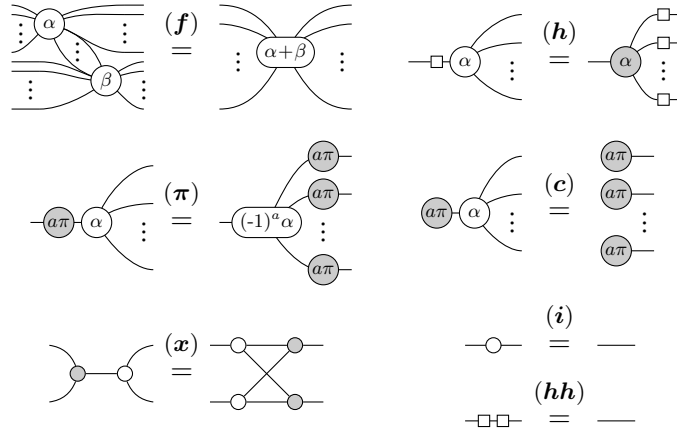
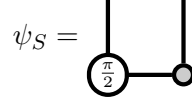


Figure 1: ZX -rules presented in [2]

(a) Let

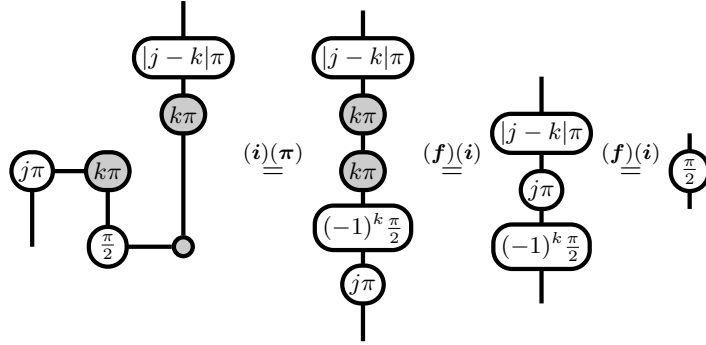


Note for $j, k \in \{0, 1\}$,

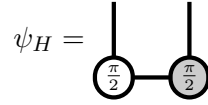
$$|j - k|\pi + j\pi + (-1)^k \frac{\pi}{2} = 2n\pi + \frac{\pi}{2}$$

for some $n \in \{0, 1\}$.

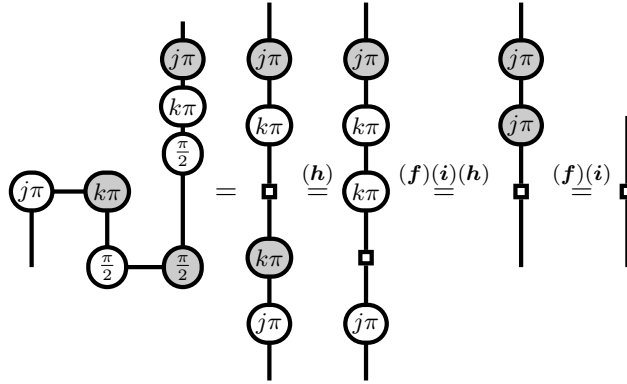
Then, using ψ_S , a 2-qubit measurement, and classically controlled Pauli X/Z gates, the S -gate is obtained as desired.



(b) Let



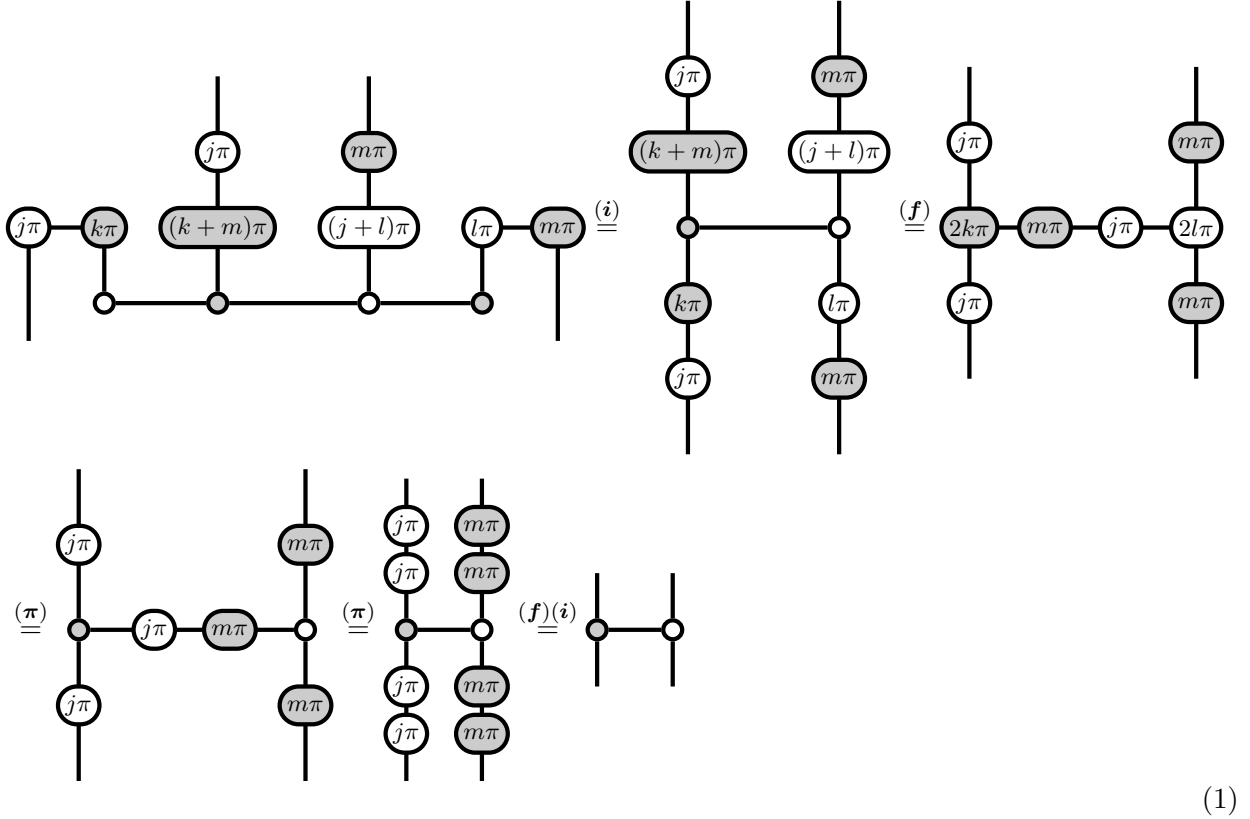
Then, also using the S -gate from part (a),



(c) Let

$$\psi_{CNOT} = \begin{array}{c} | \\ | \\ | \\ | \\ \circ - \circ - \circ - \circ \end{array}$$

so we have, using 2 two-qubit measurements, and classically controlled Pauli X/Z gates,



(d) No, since the Bell model only has $\circ \frac{n\pi}{2}$ -phase gates for some $n \in \{0, 1, 2, 3\}$, $\circ \frac{\pi}{4}$ -phase gates cannot be constructed.

First, we introduce some definitions from [1].

Definition 1.1 ([1] **Def 9.120.**). [Graph state] A graph state is a state whose ZX -diagram consists only of (0-phase) $\circ$ -spiders and H -gates where:

1. every $\circ$ -spider has exactly one output, and
2. all non-output wires connect two $\circ$ -spiders with a single H -gate.

Definition 1.2 ([1] **Def 9.103.**). [Clifford diagram] A Clifford diagram is a ZX -diagram where the phases are restricted to integer multiples of $\frac{\pi}{2}$.

Definition 1.3 ([1] **Def 9.129.**). [Clifford+T diagram] A Clifford+T diagram is a ZX -diagram where the phases are restricted to integer multiples of $\frac{\pi}{4}$.

Definition 1.4 ([1]). [Local Clifford unitaries] Local Clifford unitaries are unitaries expressible as parallel compositions of single-system Clifford diagrams.

Proposition 1.5 ([1] **Prop 9.122.**). Every Clifford diagram can be turned into a Clifford state via process-state duality, then transformed using ZX -calculus into a graph state, followed by local Clifford unitaries.

Corollary 1.6. A Clifford diagram is an extension of a graph state which can contain integer multiples of $\frac{\pi}{2}$ -phase spiders.

Corollary 1.7. A Clifford+T diagram is an extension of a Clifford diagram which can contain integer multiples of $\frac{\pi}{4}$ -phase spiders.

Theorem 1.8 ([1] **Theorem 9.130.**). Clifford+T diagrams are *approximately universal*. That is, any linear map can be approximated up to arbitrary precision by a Clifford+T diagram.

Definition 1.9 ([1] **Section 12.3.1.**). [CZ -gate] The CZ -gate can be easily constructed using a CNOT-gate and two Hadamard gates.

$$CZ - gate := D \quad \begin{array}{c} | \\ \circ \\ | \end{array} \text{---} \begin{array}{c} | \\ \square \\ | \end{array} \text{---} \begin{array}{c} | \\ \circ \\ | \end{array} \stackrel{(h)}{=} D \quad \begin{array}{c} | \\ \circ \\ | \end{array} \text{---} \begin{array}{c} | \\ \circ \\ | \end{array} \begin{array}{c} \square \\ \square \end{array}$$

Definition 1.10 ([1] **Section 12.3.1.**). Graph states can be realised by using a simple circuit as follows:

1. Prepare n qubits in the $\circ$ -state.
2. To introduce an edge between the i -th and j -th qubit, apply a CZ -gate.

Proposition 1.11. The Bell model is universal for Clifford diagrams.

Proof. Using Proposition 1.5, any Clifford diagram can be transformed to a graph state with local Clifford unitaries. Then, from Definition 1.10, the Bell model has the required CNOT-gates, Hadamard gates, and Z -basis states (converted to X -basis states using the Hadamard) to realise a graph state, along with the local Clifford unitaries. $\square$

However, Clifford diagrams are not known to be *approximately universal*. We then require the T -gate for Clifford+T diagrams following Corollary 1.7 to obtain *approximate universality* as in Theorem 1.8.

$$T := \begin{array}{c} | \\ \circ \\ | \end{array} \begin{array}{c} \pi \\ 4 \end{array}$$

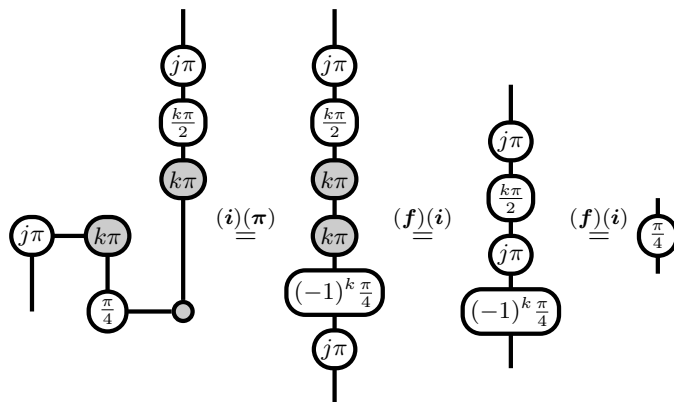
Note for $k \in \{0, 1\}$,

$$\frac{k\pi}{2} + (-1)^k \frac{\pi}{4} = \frac{\pi}{4}$$

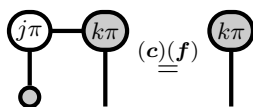
Let

$$\psi_T = \begin{array}{c} | \\ \circ \\ | \end{array} \text{---} \begin{array}{c} | \\ \circ \\ | \end{array} \begin{array}{c} \pi \\ 4 \end{array}$$

Then we can use a two-qubit measurement, and classically controlled S and Pauli X/Z gates to have



(e) It is just as powerful as before. Simply use a two-qubit measurement on the Z -basis state as follows to obtain the equivalent Z -basis measurement. The classical value of the first qubit (j) is essentially discarded.



Question 2

References

- [1] Bob Coecke and Aleks Kissinger. *Picturing Quantum Processes*. Cambridge University Press, 2017.
- [2] Aleks Kissinger and John van de Wetering. Universal mbqc with generalised parity-phase interactions and pauli measurements. *Quantum*, 3:134, April 2019.